R13

Code No: 115AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Define parse tree and provide an example.	[2]
b)	Give an example of ambiguous grammer.	[3]
c)	What do you mean by short-circuit evaluation?	[2]
d)	What is an associative array?	[3]
e)	Define sub program.	[2]
f)	What are the design issues of sub-programs?	[3]
g)	What is a semaphore?	[2]
h)	Define concurrency. State the need for concurrency.	[3]
i)	What are the features of Haskell?	[2]
j)	Compare functional languages and procedural languages.	[3]

PART - B

(50 Marks)

2.a) A programming language can be compiled or interpreted. Give relative advantages and disadvantages of compilation and interpretation. Give examples of compiled and interpreted languages.
b) In what way do operational semantics differ from denotational semantics? Explain. [5+5]

OR

3.a) List and explain the reasons for studying concepts of programming languages.
b) What purpose do predicates serve in attribute grammar? Give illustrations. [5+5]

4.a) Define Data Type. Explain in detail various primitive data types.
b) Explain the compound statements.
c) Difference between pointer and references. [4+3+3]

OR

5.a) Discuss the variable initialization and variable definition.
b) Explain the design issues for all the iterative control statements. [5+5]

6.a) Give an overview of relational and problem expressions.
b) Discuss the scope and lifetime of the variable. [5+5]

OR

7.a) Discuss the design issues of subprograms.
b) What do you mean by local referencing environment? Explain. [5+5]

8.a) Explain how Binary semaphore provides the solution to the competition synchronization problem with example pseudo-code.
b) What is exception handling? Explain exception handling in Java. [5+5]

OR

9.a) Give an overview of Java threading.
b) What are the basic elements of prolog? Explain. [5+5]

10.a) Write down the features of functional and imperative languages.
b) Explain bindings and scope in python. [5+5]

OR

11.a) Explain the fundamentals of Functional Programming language.
b) List and explain the common characteristics of the scripting languages. [5+5]

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